National University of computer and emerging scienc

Computer Organization and Assembly Language (EL-2003)

Semester: Fall 2026
Section: BAI-3A
Course Instructor: Sadaf Zehra

LAB # 02

LAB TASKS

Task 1: The Class Attendance Register (Debugging Practice)

Your friend is writing a small MASM program to store an attendance count for a class, but the program is not showing the correct value when it runs. He has already written the code and asked you to help him find the mistake.

```asm
INCLUDE Irvine32.inc

.data
attendance BYTE 0          ; total students marked present
increment  BYTE 1

.code
main PROC
     mov cl, 5              ; 5 students to mark present
     mov al, 0

L1:
     dec cl
     jz  Done               ; BUG: exits before adding the 5th
     add al, increment
     jmp L1

Done:
     mov attendance, al     ; ends up storing 4 instead of 5
     call DumpRegs
     exit
main ENDP
END main
```

- Open the given program in Visual Studio and set a breakpoint at the first line of the .code section.
- Use Step Into (F11) to execute the program one line at a time.
- Observe the Registers window after each step and note down the value of AX before and after each instruction.
- Identify the line where the value becomes incorrect and correct the mistake.

- Take a screenshot of the Registers window showing the corrected value and attach it with your solution.

Task 2: Designing a Student ID Card (Identifiers and Comments)

You have been asked to design the data section of a program that will later be used to store details for a Student ID card system. For now, you only need to declare the variables (no calculations yet).

- Write valid identifiers for: student roll number, student name (a single character grade is fine for now), and semester number.
- Make sure every identifier follows the naming rules discussed in the lab (starts with a letter, no reserved words, etc.).
- Add a single-line comment next to each variable explaining what it stores.
- Add one block (multi-line) comment at the top of your .data section describing the overall purpose of the program.

Task 3: The Grocery Store Billing Counter (Data Definition Statements)

A small grocery store owner wants a simple program that can hold the details of one item being billed at the counter. He needs different amounts of memory for different pieces of information.

- Declare a variable for the item code that only needs a small number (0-255).
- Declare a variable for the item price that can hold a larger number than the item code.
- Declare a variable for the total daily sales, which can become a very large number.
- Declare a variable to store the item name as a string, ending with a null character.
- For each variable, choose the correct intrinsic data type (BYTE, WORD, DWORD, etc.) and justify your choice in one line.

Task 4: Tracking Temperature Changes (Signed vs Unsigned Data)

A weather monitoring station records the change in temperature every hour compared to the previous hour. This change can be positive (temperature rising) or negative (temperature falling).

- Decide whether you should use BYTE or SBYTE to store the hourly temperature change, and explain why.
- Define three variables representing the temperature change for three consecutive hours (one rise, one fall, one no change).
- Write the valid range of values that your chosen data type can store.

- Run the program, add the variables to the Watch window, and verify the values match what you defined.

Task 5: Fixing the Library Book Counter (Combined Debugging Task)

The university library uses a small assembly program to keep count of how many books are issued in a day. The librarian says the final count displayed is always one less than the actual number of books issued, and she needs your help before the end of the day.

```asm
INCLUDE Irvine32.inc

.data
booksIssuedToday DWORD 0      ; DWORD: could grow large over time
bookIncrement    DWORD 1

.code
main PROC
    mov ecx, 10               ; 10 books issued today (example)
    mov eax, 0

CountLoop:
    cmp ecx, 0
    je  Finished
    dec ecx                   ; BUG: decrements before adding, so
    add eax, bookIncrement    ; last book gets skipped from the count
    jmp CountLoop

Finished:
    mov booksIssuedToday, eax  ; stores 9 instead of 10
    call DumpRegs
    exit
main ENDP
END main
```

- Identify the directives, identifiers, and data definition statements used in the given program.
- Set a breakpoint right before the counter is updated and use Step Over (F10) to move through the counting logic.
- Use Single Stepping to observe how the counter variable changes after each book is issued.
- Find the line causing the counter to be one less than expected and fix it.

- Write a short comment above your fix explaining what was wrong and how you corrected it.